

WU Hao

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EDUCATION

Beijing Institute of Technology 985 211 Double 1st-Class Bachelor of Science in Mechatronical Engineering (Robotics) Total GPA: 2.9/4.0 Major in Mechatronical Electronic Engineering of Robotics (changed) GPA: 3.5/4.0 Initially majored in Intelligent Electromechanical System GRE Score: Verbal Reasoning: 159/80% Quantitative Reasoning: 169/96% AW: 3.0/16% Total: 328/340	Sep 2019 – Jun 2024
The University of Nottingham Ningbo China QS Top 100 Electrical Engineering Master by Research (MRes)	Sep 2025 – Present

RESEARCH EXPERIENCE

Remote Research Internship in Embodied Intelligence Prof. Zhibo Pang's Research Group, Peking University Remote Intern • Investigated atomic task decomposition in long-horizon robotic tasks, designed corresponding experiments, and proposed a force-aware cutting primitive task.	Apr 2026 – July 2026 Ningbo
High-Precision Angle Error Compensation Techniques for Resolver-Based Position Sensing in PMSM Applications Master degree Topic • IEEE CIEEC 2026 Conference paper first-author. • A first-author manuscript has been submitted to IEEE Sensors Journal (JCR Q2). • One Chinese invention patent application has been filed and officially accepted. (202610975626.7)	Sep 2025 – Present Ningbo
Organoid Intelligence, at Eastern Institute of Technology in Academician JIN Dayong Group Core Member & Data Analysis Engineer • Analysis of various types of data in the direction of organoid intelligence.	Mar 2025 – Sep 2025 Ningbo
UAV and UGV Delivery Collaboration in Low-altitude Economy, at PolyU AAE AIMS Group UGV Project Leader • Mechanical design and manufacturing of UGV. • Electronic control design.	Sep 2024 – Mar 2025 Hong Kong
Research of Quadrotor UAVs Path Planning, at Beijing Institute of Technology Graduation Project • Enhanced UAV obstacle detection and avoidance using optimal A* Algorithm. • Made use of skills such as SIGMA Free to achieve predictive experiment results.	Jan 2024 – Jun 2024 Beijing
Multi-Modal Quadruped Robot Core Participant • Was responsible for remote control and mechanical structure debugging.	Sep 2023 – Jan 2024 Beijing
Cloud Shaped Floating Intelligent Companion Robot Project Leader • Was responsible for mechanical structure debugging. • Utilized STM32 for object detection, employed Bluetooth beacons for ranging and positioning. • Contributed to programming on the Raspberry Pi platform. • Project report and post.	Feb 2023 – Jul 2023 Beijing
Image Recognition of Specific Ancient Coin Text Type Project Leader	Dec 2022 – Mar 2023 Beijing

- Developed recognition of different versions of characters for Daguan Tongbao (Ten Cents).
- Proposed the project ideas and followed up, including literature research, market investigation, mentors identifying and recruiting partners, patent application, etc.
- Participated in the image collection, data set establishment, implementation of YOLOv5 algorithm, and Android app production.
- Acquired skills such as image recognition, application of YOLOv5 algorithm, etc.
- Joined Beijing Institute of Technology “Challenge Cup” Contest and got the **third award**.

Fall Prediction Based on Plantar Pressure and Attitude Signal Oct 2022 – Mar 2023

Participant Beijing

- Participated in the master students’ graduation project, patent disclosure writing, national invention patent application, and the plantar pressure detection experiment and data set establishment on this basis.

All-Terrain Spherical Robot with Deformable Structure Aug 2022 – Sep 2022

Participant Beijing

- Assisted in the design of deformation structure of the variable spherical robot and completed the structure part of the PowerPoint slides’ presentation.

The Research of 4G Remote Control Vehicle Project Dec 2021 – Apr 2022

Project Leader Beijing

- Made mechanical and electrical structure design: upgraded the vehicle door bridges, coupling, and counterweight.
- Achieved the **4G remote control** within 100 ms through ESP8266 and 4G image transmission within 500 ms through online video monitoring platform; tested the selection of different motors.
- Achieved tasks such as food delivery on complex roads through more than 50 tests.

Computational & Theoretical Chemistry Energetic

Energetic Materials Member

- **Published a SCI paper** (“Computational and Theoretical Chemistry” *Energetic Windmill: Computational Insight into Guanidine-based Nitroazole-substituted Compounds as Energetic Windmill*) as the 4th author.

WORK EXPERIENCE & EXTRACURRICULAR ACTIVITIES

Eastern Institute of Technology Academician JIN Dayong Group Mar 2025 – Sep 2025

Visiting Student & Part-Time Engineer Ningbo

- Organoid Intelligence.

Research Group of Autonomous and Interactive Mobile Systems (AIMS) at PolyU Sep 2024 – Feb 2025

Research Technical Assistant Hong Kong

- UAVs Vision, UAV and UGV delivery collaboration.

Lab of Intelligent Unmanned Systems at Beijing Institute of Technology Jun 2024 – Aug 2024

Research Assistant Beijing

- UAV Path Planning.

Beijing Bowen Kangtai Cultural Relics Company Apr 2023 – Jun 2024

Chief Numismatist Beijing & Xuzhou

- Served as a Chief Numismatist for Chinese Ancient Coins.
- Worked as a Chief Numismatic Advisor.

International Ballistics Society Feb 2022 – Present

Student Member

Siyuan Project Oct 2019 – Sep 2023

Team Leader Beijing & Ordos

- Organized and planned dozens of recruiting and cultural activities, including one marketing event with more than 3000 participants.
- Produced 3 newsletters, wrote more than 45,000 words articles, and collected more than 5GB audio and video data.
- Won one **university-level first prize for a team project, one college-level first prize for a team project, one university-level second prize for a team project, and two university-level advanced individual titles**.

The 32nd Chinese National Chemistry Olympics Competition (CChO) Sep 2018 – Dec 2018

Bronze Prize Winner Beijing & Ji-Nan

- Represented Inner Mongolia Autonomous Region to participate in the 32nd China Chemistry Olympiad and won the third prize (bronze medal) at the national level, and **obtained many guaranteed places in 985**

universities.

HONORS & AWARDS

Bronze Medal, The 32nd Chinese National Chemistry Olympics Competition	Dec 2018
Bronze Prize, Beijing Institute of Technology "Challenge Cup" Contest	July 2022
College-level Subject Competition Excellent Mention	Sep 2020, Oct 2020, Apr 2022 & Nov 2022
Honor of "Advanced Individual Title", Beijing Institute of Technology	Oct 2020, Nov 2020 & May 2021
College-level Outstanding Group Award at the BIT	May 2020, Sep 2020 & Nov 2020
First Prize, College-level "Century Cup" Competition at the BIT	July 2020

OTHER

- **Skills:** Microsoft Office, C, MATLAB, Raspberry Pi, SolidWorks, YOLO, Arduino, SIGMA Free, Isaac sim etc.
- **Language Proficiency:** Mandarin (native), English (well).
- **Interests:** Collecting ancient Chinese coins, organizing students' activities, broadcasting, sports.